Blueprint for Implementing AI

Underpin and amplify your AI capabilities with best practices from MuleSoft.

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Introduction

AI creates visions of groundbreaking possibilities and innovations from healthcare to finance, retail, and manufacturing. With **86% of IT leaders** anticipating that generative AI will strongly influence their digital transformation strategies, it's clear that most are expecting AI to drive monumental change across their organizations.[1]

The possibilities with AI today are vast. Generative AI and predictive analytics have allowed us to personalize user experiences at an unprecedented scale, enhance decision-making processes, and automate repetitive tasks across lines of business. Now, with the emergence of capabilities like autonomous agents, apps can be built to understand complex instructions, take action on the user's behalf, and even recall past conversations.

Capabilities like these have the potential to revolutionize not just marketing, sales, and service, but also logistics, finance, HR, and beyond. Think of it as giving every employee a personalized digital helper who can respond to more than a single prompt. They can take directions in natural language, create a list of next actions, and perform tasks independently so teams have more time to do what humans do best.

And while AI sparks boundless imagination and potential, certain realities must be confronted first.

Today's IT ecosystems weren't designed to handle building and managing AI apps at scale. To deploy intelligent automation with autonomous agents, you must design an IT ecosystem to be able to handle the complexities of AI – and then extend its reach across your organization.

[1] Salesforce State of IT, July 2023

What is an AI app?

There's a lot of information about what an AI app is and how you should build one. Before we delve into the specifics of setting up your IT infrastructure for AI, it's important to understand what an AI application consists of so you can properly strategize to address each component.

We've boiled AI apps down to their most essential components: the data, the user interface, the models, and the outcomes. As you continue your AI journey, keep these elements at the forefront of your strategy to ensure the best possible results from your AI initiatives.

Data

Data fuels AI by providing the context and information necessary for making informed decisions. High-quality, diverse, and relevant data is essential for developing accurate and useful AI solutions – and access to that data empowers AI models with the best raw material they can have.

User interface (UI)

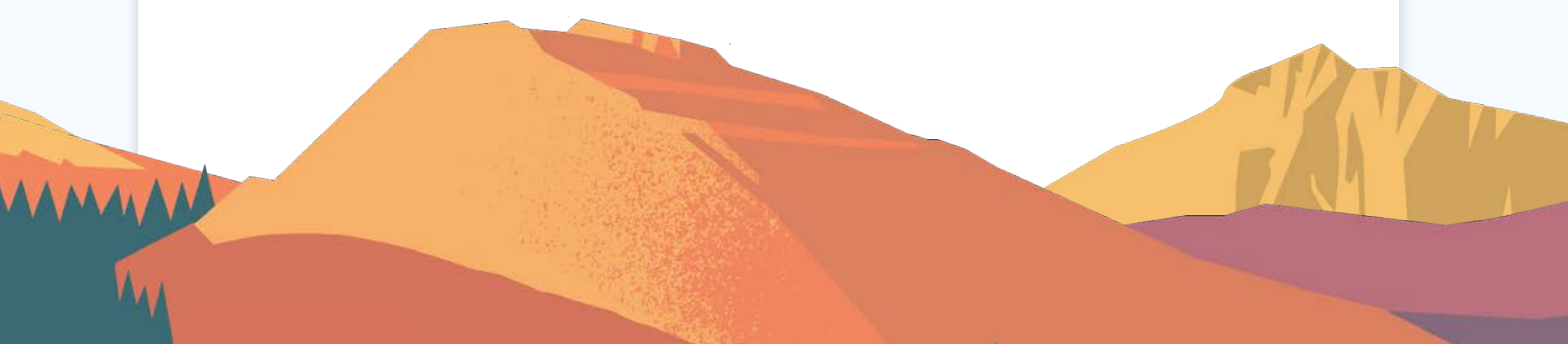
The UI is the gateway for users to interact with AI-powered systems, whether they are customers seeking assistance or employees accessing internal tools. A well-designed UI makes it easier for users to engage effortlessly with AI and make the most of their capabilities.

Models

At the heart of AI apps lie the models, encompassing the algorithms, techniques, and architectures that enable AI systems to process data, generate insights, and develop a plan of action. In the case of large language models (LLMs), deep learning techniques are used to understand and generate human-like text. However, to maintain trust and compliance, organizations must be able to make their data available to these models without compromising privacy or security.

Outcomes

In the end, every AI app must somehow improve the end user's experience. Whether fetching or updating data in real time, generating dynamic action plans, or executing tasks autonomously, AI apps must deliver tangible value for the user through more relevant answers, automated workflows, and actionable plans.



These elements come together to deliver AI apps that create value across your organization. Discovering and implementing these elements might sound like a big undertaking, but luckily you don't have to do it on your own. The Salesforce Platform provides a head start for your AI journey, bringing data, UI, models, and outcomes together for an effective AI foundation.

What makes Salesforce the #1 AI CRM?

The Salesforce Platform brings apps and data together on one integrated platform to deliver AI-powered experiences across customer relationship management (CRM) and beyond so organizations can automate and action insights across every application.

MuleSoft, Data Cloud, and Salesforce's predictive and generative AI capabilities are key components to helping organizations integrate, harmonize, and activate all of their data securely to deliver next-generation AI initiatives.

Jumpstarting your AI journey

Any time you start a complex digital project, there's no benefit to starting from zero if you don't have to. The Salesforce Platform offers a head start to AI app development by embedding data, AI, CRM, and security into one comprehensive platform. It provides IT, admins, and developers with an adaptable AI platform for rapidly (and securely) developing generative AI apps and automation.

The Salesforce Platform helps organizations hit the ground running with their AI projects with out-of-the-box functionalities that make it easier to implement AI.

For example, Salesforce houses critical customer **data** needed for AI. With Data Cloud, organizations can unify and augment data from all their CRM instances, providing valuable insights and a fuller understanding of their clientele.

Salesforce also enables intuitive interaction with and actioning of AI capabilities with Agentforce as a **user-friendly interface**. Agentforce is a set of tools to help helps users create and customize AI agents – both autonomous and assistive – without requiring deep technical expertise.

Meanwhile, Salesforce's Trust Layer offers native integration with **models** to ensure the safety and security of sensitive information. Organizations can use pre-trained models or bring in their own – whichever approach is needed to meet their specific business needs and requirements.

Whatever your objective is for your AI project, Salesforce lets you produce high-impact **outcomes** across domains like sales, service, and marketing. Plus, the integration and automation capabilities from MuleSoft help you action outcomes to any system or application across your organization.

From helping organizations manage customer relationships, forecasting demand, and providing real-time service chat support, you can use the Salesforce Platform to drive business growth and increase customer satisfaction. By embedding data, UI, models, and outcomes within the Salesforce Platform, Salesforce equips organizations with the tools and capabilities needed to get up and running quickly with AI.

Barriers to AI adoption

The Salesforce Platform greatly accelerates AI endeavors across CRM. However, activating data, UIs, models, and outcomes outside of CRM – that reach across any application or workflow in your organization – requires added attention.

Although Salesforce includes access to customer data in CRM, organizations likely have a lot of other data scattered across hundreds of external systems. Shockingly, **the average enterprise uses nearly 1,000 systems**. And often, connecting all that data is a top barrier to AI implementation – **95% of decision-makers** say their main roadblock to AI adoption is integration.[2]

Disconnected systems frequently result in poor data access, quality, and structure inconsistencies. Employees may find navigating and interacting with multiple user interfaces for different applications challenging. Additionally, managing and maintaining numerous third-party apps or user interfaces can be burdensome for IT departments, as updates, patches, and security fixes must be applied to each application individually, increasing the risk of errors, vulnerabilities, and downtime.

Meanwhile, the rapid proliferation of AI models (from just 30 commercially significant ones a year ago to over 500,000 unique models available today) poses significant challenges for AI adoption for the average enterprise.[3]

With such a vast array of AI models to choose from, many struggle to navigate the options and select the most suitable model for their specific use case. Integrating multiple AI models into existing systems or building your own model can be complex and time-consuming, with each model having its own requirements, dependencies, and compatibility issues.

Lastly, many AI use cases will need the ability to extend beyond CRM. If a desired outcome lives outside of Salesforce – for instance, your ERP – integration is needed to start an action in one system and continue actioning outcomes in another.

For example, imagine an autonomous agent app that begins its actions in CRM by analyzing customer data, sales trends, and interactions to predict future sales. It then takes these predictions to the ERP system to automatically adjust procurement, inventory levels, and financial forecasts accordingly. This kind of seamless integration between systems is necessary to enable a fully end-to-end, hands-free sequence of actions for your AI apps.

[2] MuleSoft 2024 Connectivity Benchmark Report
[3] Hugging Face

Four-step blueprint for implementing AI

As a critical component of the Salesforce Platform, MuleSoft extends the platform's capabilities to systems beyond your CRM so you can realize greater success with your AI initiatives. To help direct you on your journey, we've created a holistic four-step approach to AI success, using APIs to help you unlock your data and automate actions across every system and workflow in your ecosystem.

1 Connect all of your data for seamless access

Embedded in the Salesforce Platform, Data Cloud connects and harmonizes customer data in Salesforce, eliminating data silos that would traditionally prevent AI applications from accessing relevant data. Data Cloud offers native connectivity for all first-party data in Salesforce and major cloud storage solutions like Amazon S3, Google Cloud, and Microsoft Azure. It also allows for zero-copy data sharing and federation for major data lakes and warehouses including Snowflake, Databricks, Google BigQuery, and select SaaS applications.

Although your CRM data is valuable, it only provides a snapshot of one facet of your organization. To build truly versatile AI apps, you also need to be able to incorporate data about products, transactions, consumer behavior, and more. In addition to CRM data, MuleSoft seamlessly unlocks data from other systems, including ERP, HR, Finance, and homegrown or legacy systems to ensure the necessary information is accessible and available for your AI applications.

Establishing seamless data flows from these systems allows you to feed your AI algorithms with the most accurate, accurate, and up-to-date information – a process known as grounding. Grounding helps ensure the AI apps you build are connected to reality in a meaningful way. It involves connecting AI's abstract concepts and algorithms to diverse, correct, and current information from the real world. This allows AI to understand and interact with the environment in a relevant and appropriate manner. Without grounding, your AI systems might operate on outdated or irrelevant information, leading to errors and unhelpful interactions.

Extending Data Cloud capabilities with MuleSoft

If you're like most companies, you have essential data residing within industry-specific systems such as FIS and Epic, across legacy platforms like SAP and Oracle, in custom-built solutions like mainframe systems, and other cloud-based applications.

Using APIs to connect disparate systems and applications, MuleSoft helps bring all of your data – no matter where it resides – into Data Cloud so you can ground your AI in diverse datasets and fuel more comprehensive outcomes across any line of business.

2 Compose your data for specific business functions and user interfaces

Unlocking your data and making it accessible serves the same role as the first step to cooking a gourmet meal. The chef gathers all the necessary ingredients in the kitchen, making the ingredients accessible for when they're needed in the cooking process. However, without the right tools, even the best chef would have difficulty turning those ingredients into the ultimate gourmet dish.

Similarly, after unlocking your data to make it accessible, it's critical to have an architecture that lets you transform and serve that data to its destination – the user interface. By composing your data in an API-led architecture and taking a three-layered approach to organizing your IT APIs and assets, you enable the most efficient use of your data for the specific business applications you've chosen for it, including AI apps. MuleSoft's Anypoint Platform is the leading platform for API-led connectivity.

Three-layered connectivity: System, Process, and Experience APIs

System APIs

These APIs unlock the data stored within an organization's core systems of record. Examples of critical systems from which APIs can unlock data include ERP, customer and billing systems, and proprietary databases.

Process APIs

By composing data surfaced from System APIs, Process APIs aggregate and shape data for specific business purposes, such as gathering customer or employee information or order details. Process APIs facilitate combining data and orchestrating multiple System APIs for targeted business objectives – for example, creating a 360-degree view of the customer, managing order fulfillment, or tracking shipment status.

Experience APIs

Experience APIs expose data and logic to applications or interfaces. They provide a business context for the data and processes unlocked and established by System and Process APIs. Experience APIs ensure that the exposed data is consumable by its intended audience, such as the AI user interface.

3 Control your data to enable trusted AI

Once your data has been unlocked and efficiently applied to an AI project, you need a way to ensure reliability and security. You must be able to govern and secure your APIs and the data used in your AI project to ensure sensitive information doesn't become exposed to unauthorized individuals or public AI models.

You can effectively control access and prevent sensitive information from being shared with LLMs using an API management solution like MuleSoft, which provides several key benefits:

Continued reliability: Once your AI app is operational, it needs continuous access to data to power consistently useful output. MuleSoft APIs provide reliable integrations across systems, processes, and experiences, so AI apps like autonomous agents have consistent access to the data they need when they need it.

Governance and security: API governance ensures that APIs are managed effectively by setting access controls, authentication mechanisms, and usage policies. It also allows you to monitor API performance, usage patterns, and potential vulnerabilities to ensure ongoing security.

Sensitive information protection: AI often involves sensitive data, including customer records, financial transactions, health information, and so on. API management helps prevent accidental or malicious leaks that could have severe consequences.

Organizational reputation and compliance: Compliance with data protection laws (like GDPR) is non-negotiable. API management helps centralize and control privacy and security requirements.

4 Action AI recommendations across your business

The framework's final step is the ability to act on the insights, resolutions, or workflows recommended by your AI application to drive the best and most relevant outcomes.

The reality is that many desired outcomes and actions will exist in numerous systems that extend beyond Salesforce. While insights related to customers and marketing exist in Salesforce, workflows attached to an e-commerce process, for instance, may happen in other systems.

Suppose a customer purchases a product recommended by AI. In this case, cascading events occur across different platforms and systems. The purchase needs to be sent to the order management system and logged in the ERP. It then needs to be sent to the order fulfillment system and posted to the accounting system when the product ships.

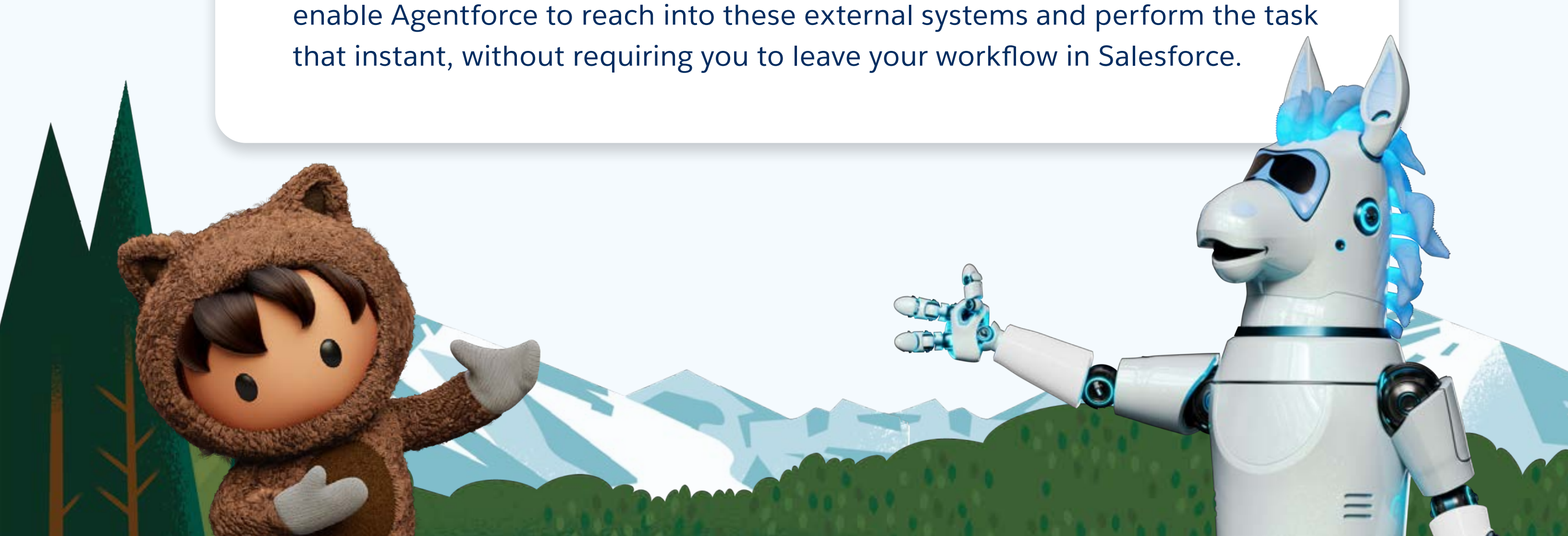
To enable a process like this, you need a way for your AI apps to access other systems and trigger the next actions based on previous events. This capability allows AI apps like autonomous agents to perform actions on your behalf, even if the data and workflows involved reside in systems outside your CRM.

That's where MuleSoft APIs and bots can facilitate real-time data movement based on events that occur throughout this process end-to-end.

Extending Agentforce capabilities with MuleSoft

Agentforce is a suite of customizable agents and tools that seamlessly integrate with an organization's data and apps. Acting as a trusted advisor and helper, it provides quick answers to the user's queries and acts on their behalf, saving them time and making them more productive. While Agentforce supports actions within Salesforce, MuleSoft gives it the ability to act outside of Salesforce as well, using APIs and robotic process automation (RPA) bots.

For example, you could ask Agentforce to update an invoice amount in your ERP or place an order in your order management system. MuleSoft would then enable Agentforce to reach into these external systems and perform the task that instant, without requiring you to leave your workflow in Salesforce.



Applying the four steps

These four steps – connect, compose, control, and action – comprise successful AI implementation. Through these four steps, you can ensure the accessibility, integration, security, and reliability of your information from as many disparate sources as possible, while empowering your AI and team to turn that data into actionable insights.

Let's explore how one company used MuleSoft to integrate its IT ecosystem and fuel AI innovation with Salesforce Platform.

AI success story: IRON MOUNTAIN®

Iron Mountain is a global information management services company that serves more than 225,000 organizations around the world, including 95% of the Fortune 1000. With a mission to protect and elevate the power of its customers' work and with a customer base of this size and scale, providing an exceptional customer experience is a top priority.

Despite having massive amounts of customer data, Iron Mountain's service representatives couldn't easily access it or use it meaningfully to address customer requests. Service reps had to switch between multiple apps and databases to gather enough data for customer inquiries. Plus, the order process was manual and information was located across numerous places, forcing them to go back and forth between systems.

Iron Mountain turned to Salesforce technology to improve service representatives' experience. It helped the company:

- **Close cases faster:** Using generative AI to automatically write personalized case replies based on past cases and the knowledge base, service reps can more quickly find the relevant information to close out cases.
- **Reduce administrative work:** Using Work Summaries, service reps can also move on from a case more quickly, thanks to Work Summaries powered by the Salesforce Platform. When a case is complete, Agentforce Service Agents automatically generate a summary based on the case data, conversation, and transcript history, reducing the administrative load that service reps previously managed.
- **Integrate workflows:** Iron Mountain uses the Salesforce Platform to pull the right data for each case. This helps generate the right responses from AI and fully integrates their service workflows directly in Service Cloud. Behind the scenes, connected data fuels an improved experience for service reps. MuleSoft brings together account, billing, and engagement data all in one place, so that orders can be automated and service reps have a holistic view of each customer.

Learn more on how Iron Mountain increases service representative productivity by unifying data.

Conclusion

AI apps like autonomous agents are set to revolutionize organizations' operations, enabling adaptive decision-making, automating workflows, and freeing employees for tasks requiring a human touch. Before building one, however, you need an IT foundation that provides unobstructed, secure access to data no matter where it resides.

MuleSoft makes this possible by removing barriers between data sources and streamlining flows between CRMs and other platforms with robust API connectivity. From gathering and shaping data to presenting it in the most powerful way possible, MuleSoft radically empowers you to achieve more with your data and embrace building with AI.

Discover

Explore how the Salesforce Platform facilitates fast AI development and automation.

Explore the Salesforce Platform

Webinar

Find out how MuleSoft and Salesforce tools work together to accelerate innovation.

Watch the webinar

Blog

Empower Agentforce to Act Across the Enterprise With MuleSoft APIs.

Read the blog

Demo

Learn how Anypoint Flex Gateway helps you effectively control and manage LLM APIs.

Watch demo

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For more information please visit Salesforce.com, or call 1-800-No-Software.